

cloud store

- stakeholders: store owner / manager, Investors, delivery workers, integrators / payment service
- Core functionality: it allows customers to easily purchase their goods from a cloud supermarket on the devices from anywhere in the world directly to their homes.
- user (system scenario): The customer first logs in to our program then begins to choose products from the site and put them in his shopping cart. When finished, they complete their order and go directly to the payment page. After payment, the store receives the orders, prepares them, and sends them with the delivery representative. Delivery is then made to the customer.
- Store Owner/manager (system scenario): Utilizes the system to maintain inventory and manage the physical flow of goods. They monitor alerts for low-stock items, using the system to view sales history and generate optimized purchase orders for suppliers.
- investors scenario (system scenario): Investors access a secure, read-only Executive Dashboard to evaluate the business's health and potential growth. Their primary interaction is pulling aggregate reports to monitor key performance indicators (KPI) such as monthly gross merchandise value (GMV) and customer metrics like acquisition cost and retention rate.

use cases, their primary actors and their description.

UC1 - Manage Digital storefront and Inventory:- The actor maintains the product catalog, updates Stock levels, monitors for low stock, and generates purchase orders (PO) to suppliers (store owner/manager).

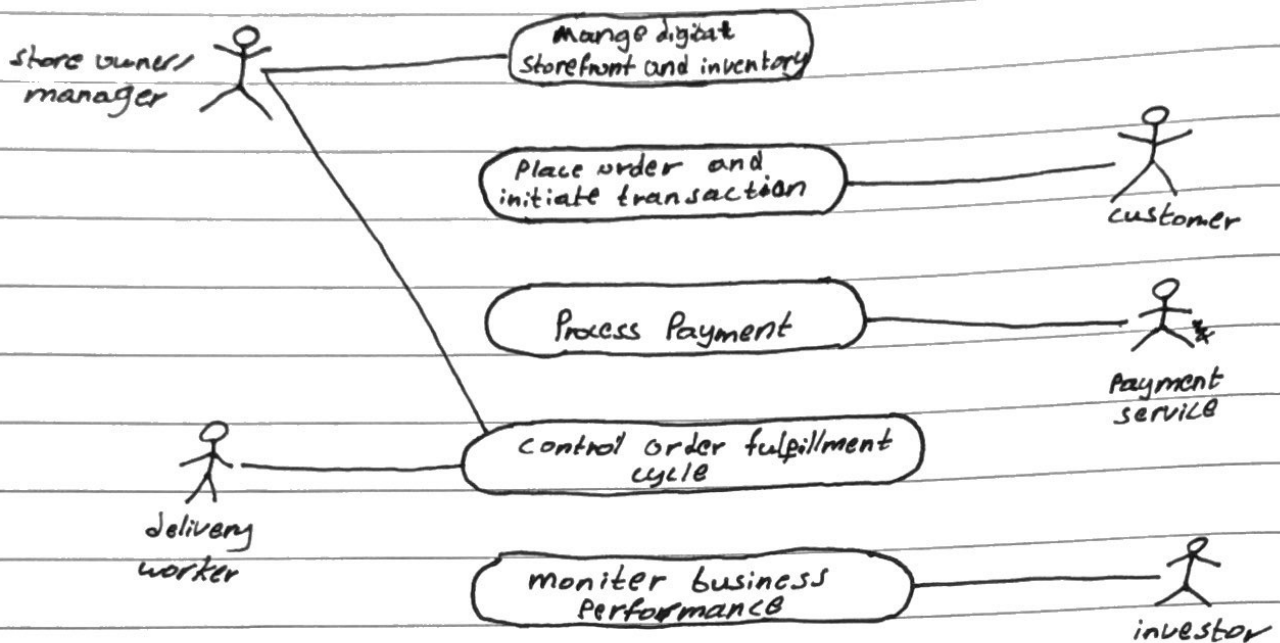
UC2 - Place order and initiate transaction:- The customer browses products, manages their cart, finalizes their order details, and submits their request to purchase which triggers the payment process (customer).

UC3 - Process payment:- the system sends the transaction request to the external service, handles the security token exchange, and receives the final success/failure confirmation to finalize the sale (Payment service).

UC4 - control order fulfillment cycle:- Manages the logistics: generating picking lists, assigning orders to Delivery Routes, and executing/confirming the final delivery status via mobile app (store owner/manager, delivery worker).

UC5 - monitor business Performance:- The investor securely accesses the system to generate detailed Executive Reports on key performance indicators (KPI) like (GMV), Retention Rate and inventory metrics for strategic analysis. (investor)

Use case diagram:



Non-function requirements -

Security - All PII and transaction data related to UC2/UC3 must be handled in strict compliance with PCI DSS and secured with up-to-date encryption protocols.

Integrability - The Payment service API (UC3) and the delivery API (UC4) must be fully documented using a formal specification (OpenAPI) to ensure stability and extensibility.

Performance - The entire transaction pipeline, from order submission (UC2) through payment confirmation (UC3), must complete in under 3 seconds.

Availability - The storefront (UC2) and payment integration (UC3) must maintain 99.9% uptime to guarantee continuous sales capability.

Scalability - The system must be able to handle a 500% increase in transaction volume over 12 months without requiring architectural re-design or degrading response time.

Auditability - All inventory and financial data must be logged, traceable and available for automated report generation (UC5) with data retention of at least 10 years.

Initial analysis :- (Objects and Classification)

Boundary objects - Interface with the outside world (actors), managing screens, forms, reports, and external APIs.

(ShoppingCartView, ProductCatalogView, PaymentAPI, DeliveryAppInterface, LoginScreen, ManagerDashboard, ExecutiveReportView).

Entity Objects - Model persistent information and core domain concepts; they are long-lived and represented the data.

(Customer, Product, Cart, Order, ~~LineItem~~ LineItem, InventoryStock, DeliveryRoute, Address, PaymentToken).

Control objects - Manage the application logic for specific use cases, orchestrating the flow between Boundary and entity objects.

(OrderPlacmentHandler, PaymentProcessor, FulfillmentManager, RouteOptimiser, ReportGenerator).